Xi'an Jiaotong-Liverpool University
西交利物浦大学

RBE211TC DYNAMIC SYSTEMS

VIBRATION OF MECHANICAL SYSTEMS – MDOF SYSTEMS (PART 1)

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Lecture Outline

1. Why SDOF models are not always sufficient
2. Examples of coupled systems in engineering and robotics
3. Equations of motion for 2DOF systems
4. Matrix form of MDOF systems
5. Physical meaning of mass, damping, and stiffness matrices
6. Coupling terms and what they imply physically
7. Brief preview of multiple natural frequencies

From Earlier Modelling to MDOF Thinking

Earlier in the module, we learned how to:

- choose suitable coordinates
- derive equations of motion for mechanical systems
- represent system behaviour mathematically

Today, we build on that foundation by focusing on:

- systems with multiple coordinates
- coupled motion
- matrix representation

Why Are SDOF Models Not Always Enough?

An SDOF model may be insufficient when:

- different parts of the system can move independently
- translation and rotation occur together
- one component influences the motion of another
- one coordinate cannot fully describe the motion

Key idea:
More independent motion coordinates lead to more degrees of freedom.

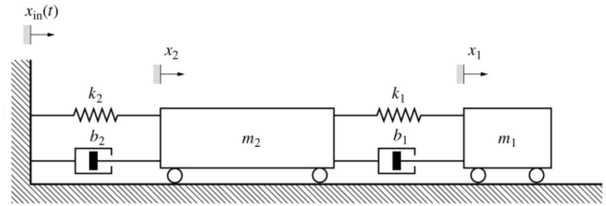
Examples of Multi-DOF Systems

Examples include:

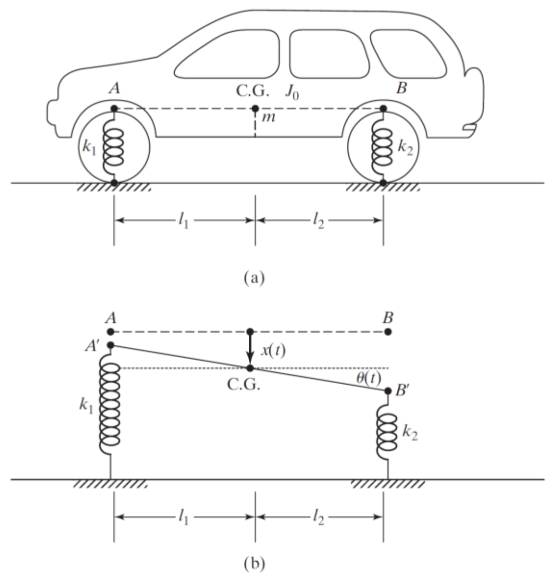
- two masses connected by springs and dampers
- a cart–pendulum system
- a vehicle bounce–pitch model
- a robotic arm with multiple links
- a flexible structure with interacting parts

Key question:

What coordinates are needed to describe the motion fully?



Two-mass system.

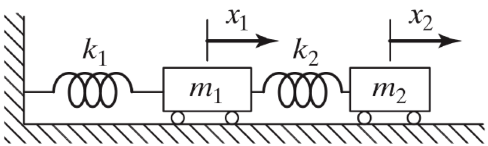


Simple vehicle bounce-pitch model.

What Is Coupled Motion?

In a coupled system:

- motion in one coordinate affects another coordinate
- the equations of motion are linked
- the system cannot be described by independent SDOF equations



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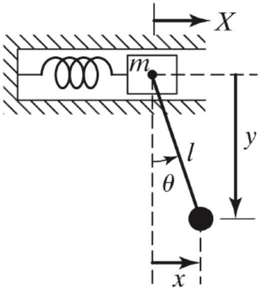
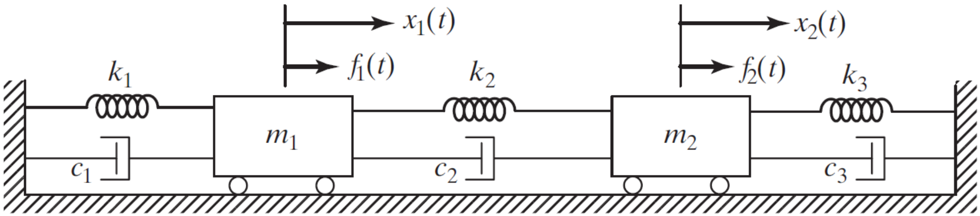


Cart-pendulum.

Physical meaning:
The coordinates do not evolve independently.

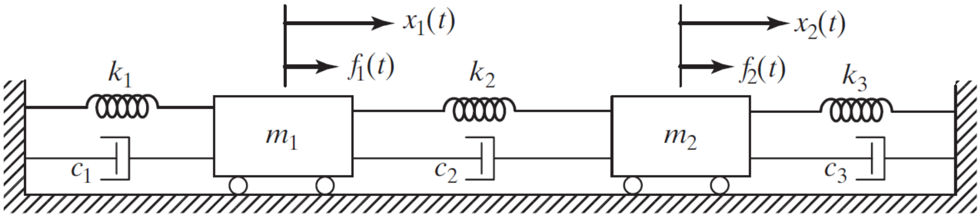
A Simple Two-Degree-of-Freedom System



Consider a 2DOF spring–mass–damper system with coordinates:

- $x_1(t)$: displacement of mass m_1
- $x_2(t)$: displacement of mass m_2

Equations of Motion of the 2DOF System

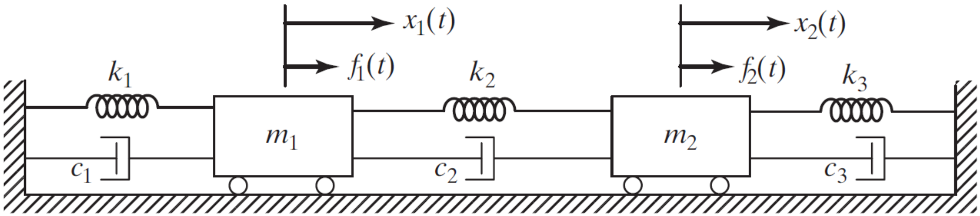


Applying Newton's second law to each mass gives:

Observation:

Each equation involves both $x_1(t)$ and $x_2(t)$.

Why Are the Equations Coupled?

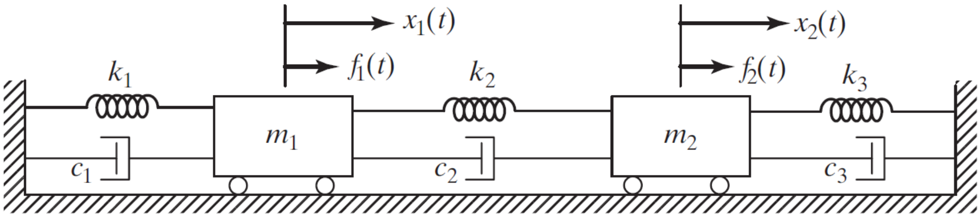


The equations are coupled because:

- x_2 appears in the equation for x_1
- x_1 appears in the equation for x_2
- similarly for $\dot{x}_1$ and $\dot{x}_2$

This happens because the two masses are connected through shared elements.

Matrix Form of an MDOF System



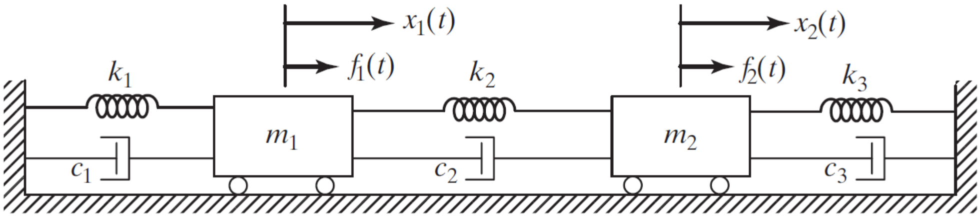
The coupled equations can be written compactly as

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{f(t)\}$$

where:

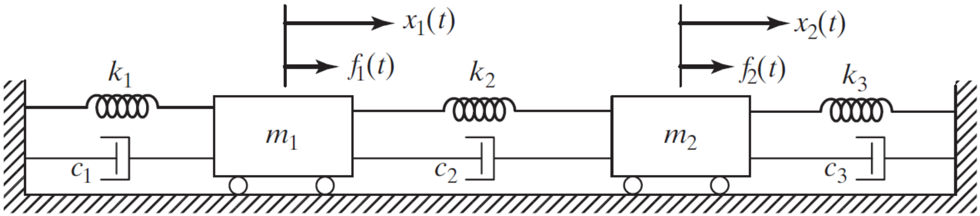
- $[M]$: mass matrix
- $[C]$: damping matrix
- $[K]$: stiffness matrix
- $\{x\}$: displacement vector
- $\{f(t)\}$: force vector

Mass, Damping, and Stiffness Matrices



For the 2DOF mass-spring-damper system:

Physical Meaning of the Mass Matrix



The mass matrix $[M]$:

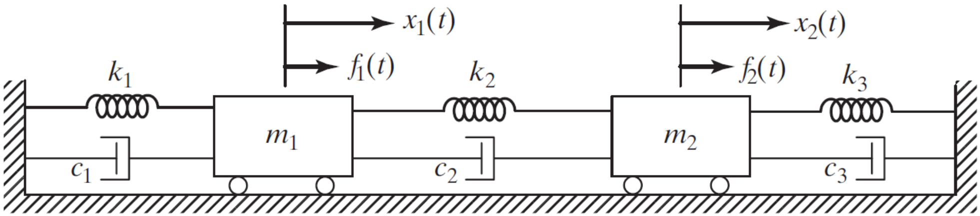
- represents inertia in each coordinate
- relates acceleration to inertial force
- is often diagonal in simple translational systems

$$[M] = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}$$

For this example:

- m_1 affects $\ddot{x}_1$
- m_2 affects $\ddot{x}_2$

Physical Meaning of the Stiffness Matrix



The stiffness matrix $[K]$:

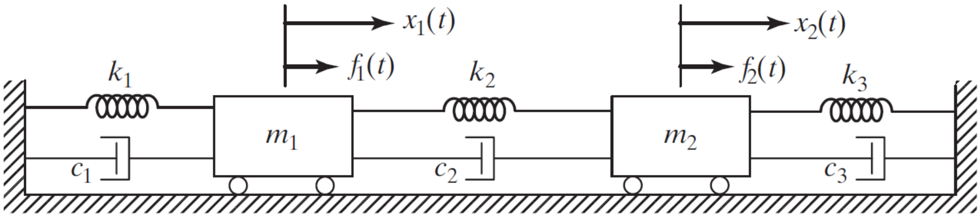
- represents restoring forces due to displacement
- diagonal terms represent direct stiffness effects
- off-diagonal terms represent coupling between coordinates

$$[K] = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix}$$

For this example:

- k_2 connects the motion of x_1 and x_2

Physical Meaning of the Damping Matrix



The damping matrix $[C]$:

- represents energy dissipation due to motion
- diagonal terms represent direct damping effects
- off-diagonal terms represent velocity coupling

$$[C] = \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 + c_3 \end{bmatrix}$$

For this example:

- c_2 causes the velocity of one mass to influence the other

Interpretation of Off-Diagonal Terms

Off-diagonal terms indicate coupling.

Examples:

- K_{12}, K_{21}: stiffness coupling
- C_{12}, C_{21}: damping coupling
- M_{12}, M_{21}: inertia coupling (in more general systems)

Key idea:
The coordinates cannot be treated as independent

Coupled and Uncoupled Systems

An uncoupled system has equations that can be written independently.

A coupled system has equations in which:

- one coordinate appears in another equation
- the motion must be solved together

Key observation:
Matrix form makes this distinction immediately visible.

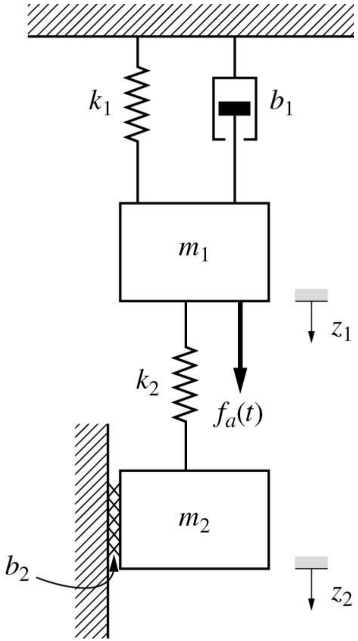
$$[M] = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \qquad [K] = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix}$$

Revisiting Earlier Multi-Coordinate Examples

From earlier modelling work:

- some mechanical systems already required more than one coordinate
- the equations were already multi-coordinate
- today we reinterpret them as MDOF systems

Question:
Where does the coupling appear in the equations?



Example 4 of Modelling Mechanical Systems

Previously, the equations of motion were obtained as:

$$\begin{array}{ll} z_1: & \boxed{\begin{array}{l} m_1\ddot{z}_1(t) + b_1\dot{z}_1(t) + (k_1 + k_2)z_1(t) - k_2 z_2(t) = f_a(t) \\ \\ m_2\ddot{z}_2(t) + b_2\dot{z}_2(t) + k_2 z_2(t) - k_2 z_1(t) = 0 \end{array}} \\ z_2: & \end{array}$$

This system can be written compactly in matrix form as:

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{Bmatrix} \ddot{z}_1 \\ \ddot{z}_2 \end{Bmatrix} + \begin{bmatrix} b_1 & 0 \\ 0 & b_2 \end{bmatrix} \begin{Bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{Bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} z_1 \\ z_2 \end{Bmatrix} = \begin{Bmatrix} f_a(t) \\ 0 \end{Bmatrix}$$

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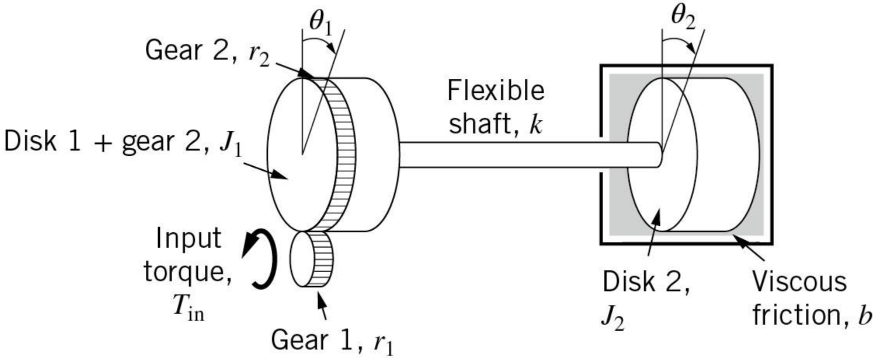


Revisiting Earlier Multi-Coordinate Examples

Example 5 of Modelling Mechanical Systems

Previously, the equations of motion were obtained as:

$$\theta_1: \quad J_1\ddot{\theta}_1 + k(\theta_1 - \theta_2) = \frac{r_2}{r_1}\tau_{in}$$

$$\theta_2: \quad J_2\ddot{\theta}_2 + b\dot{\theta}_2 + k(\theta_2 - \theta_1) = 0$$

This system can be written compactly in matrix form as:

$$\begin{bmatrix} J_1 & 0 \\ 0 & J_2 \end{bmatrix} \begin{Bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & b \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{Bmatrix} + \begin{bmatrix} k & -k \\ -k & k \end{bmatrix} \begin{Bmatrix} \theta_1 \\ \theta_2 \end{Bmatrix} = \begin{Bmatrix} \frac{r_2}{r_1}\tau_{in} \\ 0 \end{Bmatrix}$$

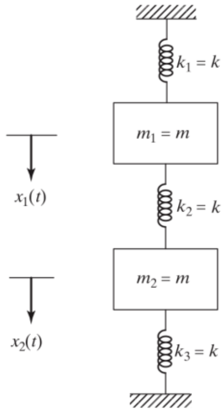
Why Do Multiple Natural Frequencies Arise?

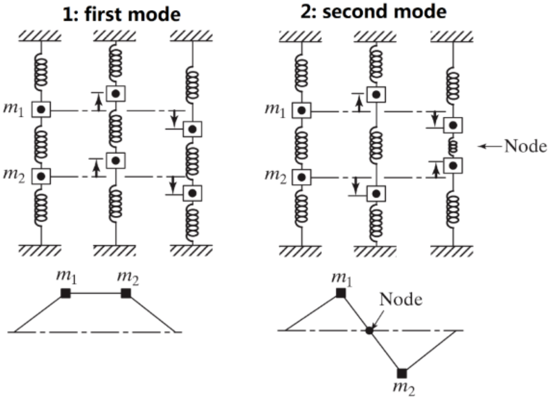
In an MDOF system:

- there is more than one independent coordinate
- the system can vibrate in more than one characteristic way
- each characteristic way is associated with a natural frequency

Key idea:

More degrees of freedom lead to more possible vibration patterns.





From Coupled Equations to Mode Shapes

If we consider undamped free vibration,

$$[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$$

then the system leads to:

- multiple natural frequencies
- multiple mode shapes
- an eigenvalue problem

Next step:

- free vibration of MDOF systems
- natural frequencies
- mode shapes

Identify the Matrices

Example

Given

$$2\ddot{x}_1 + 0.5\dot{x}_1 - 0.5\dot{x}_2 + 5x_1 - 2x_2 = f_1(t)$$

$$\ddot{x}_2 - 0.5\dot{x}_1 + 0.8\dot{x}_2 - 2x_1 + 4x_2 = f_2(t)$$

Write the system in the form

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{f(t)\}$$

Example

From the equations,

$$[M] = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[C] = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.8 \end{bmatrix}$$

$$[K] = \begin{bmatrix} 5 & -2 \\ -2 & 4 \end{bmatrix}$$

$$\{f(t)\} = \begin{Bmatrix} f_1(t) \\ f_2(t) \end{Bmatrix}$$

What If the Matrices Were Diagonal?

If

$$[M], [C], [K]$$

were all diagonal, then:

- each coordinate would behave independently
- the equations would become uncoupled
- the system would behave like separate SDOF systems

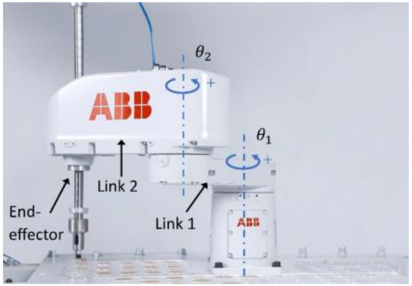
Key message:
Off-diagonal terms are the mathematical signature of coupling.

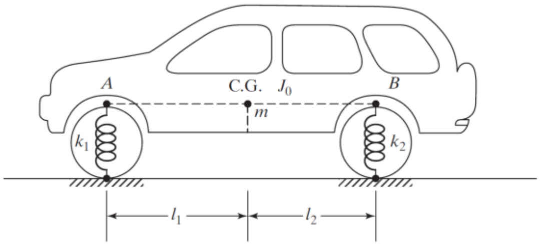
Multi-DOF Behaviour in Robotics

In robotics, MDOF behaviour appears naturally in:

- multi-link manipulators
- mobile robots with body pitch and bounce
- flexible arms and end-effectors
- anti-sway and suspended-load systems

Key idea:
motion in one part of the robot can influence another part.

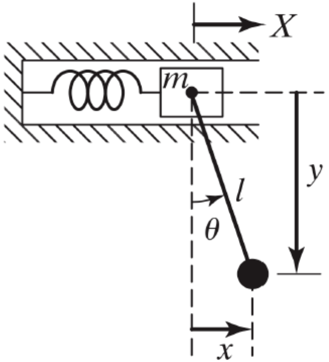




Example: Translational–Rotational Coupling

A cart–pendulum system may require:

- cart displacement $x(t)$
- pendulum angle θ



Physical meaning:

- cart motion affects pendulum motion
- pendulum motion affects cart reaction
- the coordinates are coupled

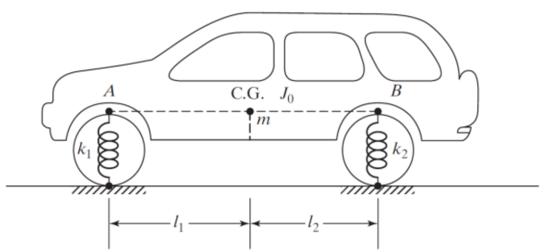
Example: A Vehicle as a Two-Degree-of-Freedom System

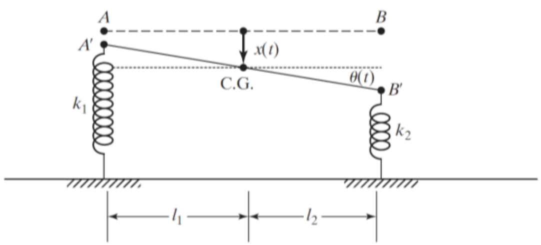
A simplified vehicle body may require:

- vertical displacement $x(t)$
- pitch angle $\theta(t)$

Why?

- road inputs at front and rear are different
- the body may bounce and rotate simultaneously





Review

- MDOF systems require multiple independent coordinates
- Coupling means the coordinates influence one another
- Equations of motion can be organized into matrix form
- $[M]$,$[C]$, and $[K]$ have clear physical meaning
- Off-diagonal terms reveal coupling
- MDOF systems can have multiple natural frequencies